

Food Vision Big

Deep Learning Food Image Classification System

TRANSFER LEARNING

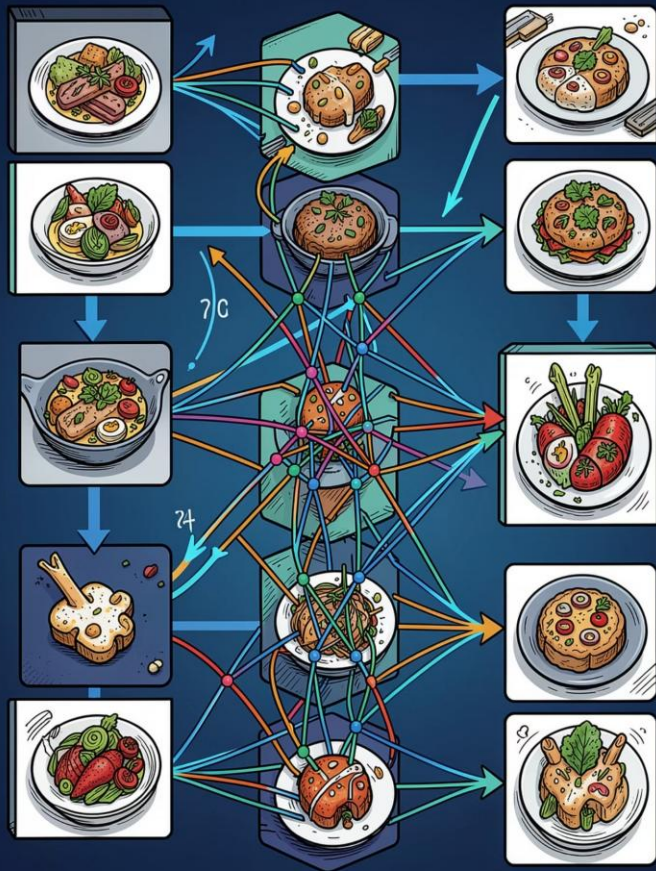
COMPUTER VISION

TENSORFLOW

DEEP LEARNING



AI Deep Learning From One's Maturation



NOOD'S:

PROJECT OVERVIEW

End-to-End Food Classification at Scale

Food Vision Big™ is a production-grade deep learning pipeline built to classify food images across 101 categories — trained on the Food101 dataset using TensorFlow and transfer learning.

101

Food Categories

75K

Training Images

25K

Test Images

101

Output Classes

Dataset & Pipeline Architecture

The pipeline leverages TensorFlow's **tf.data API** for parallel loading, automatic prefetching, and GPU-optimized batching — eliminating I/O bottlenecks during training.

- Parallel data loading across CPU cores
- Automatic prefetch to hide latency
- GPU-optimized batch processing
- Seamless TFDS integration

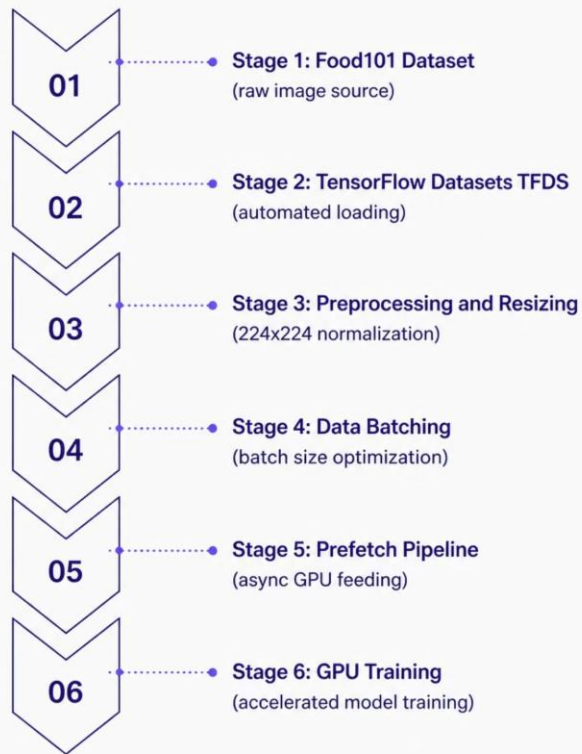
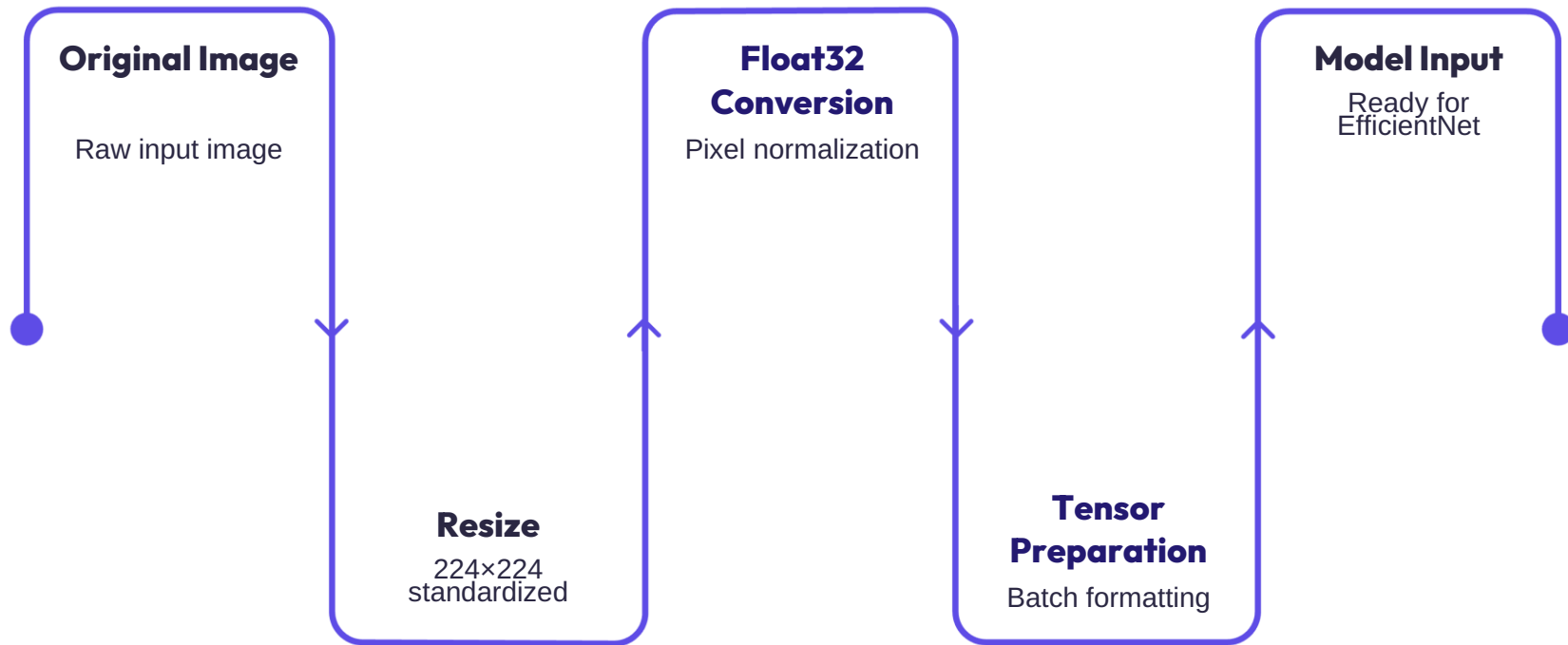


Image Preprocessing Pipeline



Every input image is resized to **224x224**, converted to **float32**, and normalized — ensuring consistent, production-ready tensors for EfficientNetB0.

TensorFlow Image Processing

Native tf.image ops for speed

Automatic Preprocessing

Built into the training pipeline

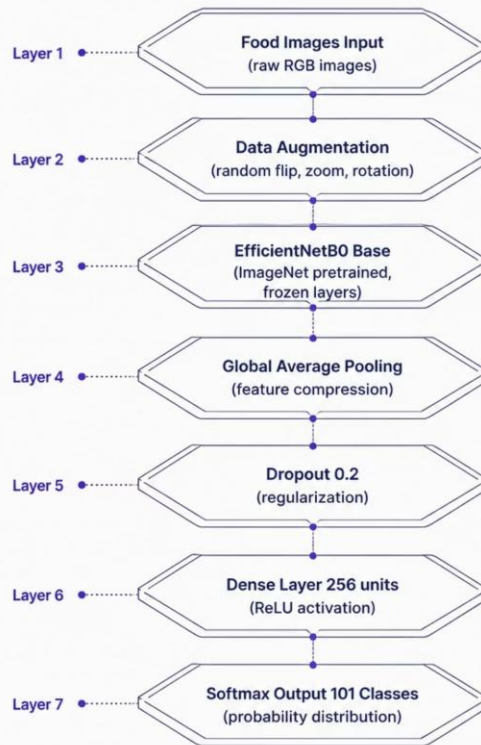
Production-Ready Flow

Zero manual intervention required

Deep Learning Model Architecture

Built on **EfficientNetB0** pretrained on ImageNet — combining transfer learning, feature extraction, and fine-tuning for high-accuracy food classification across 101 classes.

- Transfer Learning from ImageNet weights
- Feature Extraction via convolutional base
- Fine-Tuning for domain adaptation
- Softmax output across 101 food classes



Training Optimization Strategy



Mixed Precision Training

FP16 + FP32 compute for faster training and reduced GPU memory usage



TensorBoard Monitoring

Real-time visualization of accuracy, loss, and learning rate curves



EarlyStopping

Halts training when improvement stalls — prevents overfitting



Model Checkpoints

Auto-saves best weights each epoch — zero risk of losing progress



ReduceLROnPlateau

Automatically lowers learning rate when validation loss plateaus



GPU Acceleration

Full CUDA optimization via Google Colab GPU runtime

PERFORMANCE

Model Performance Dashboard



Key Results

- ~90% **validation accuracy** achieved across 101 food classes
- Clean convergence — minimal overfitting observed
- TensorBoard tracks accuracy, loss, and LR in real time
- Confusion matrix validates per-class prediction quality

📄 Performance metrics are illustrative — actual results vary by configuration and training run.

TECH STACK

Technology Stack

Programming

Python

Frameworks

TensorFlow · Keras · TensorFlow
Datasets

Libraries

NumPy · Matplotlib · TensorBoard

Hardware

Google Colab GPU · CUDA · Mixed Precision

Architecture

EfficientNetB0 · Transfer Learning · Computer Vision

APPLICATIONS

Real-World Applications



Restaurant Automation

Automated menu recognition, ordering systems, and kitchen workflow optimization



Nutrition & Health

Smart food logging, diet monitoring, and personalized healthcare nutrition insights



AI Camera Systems

Edge-deployed vision models for real-time food identification in smart environments



Mobile AI

Lightweight deployment for on-device food recognition and smart assistant integration

Food Vision Big

Deep Learning Food Classification — Built for Production

- **End-to-End Deep Learning Pipeline**
- **EfficientNetB0 Transfer Learning**
- **Mixed Precision Training + TensorBoard Monitoring**
- **101-Class Food Classification at Scale**

TENSORFLOW

KERAS

PYTHON

GOOGLE COLAB

